

Machine Learning Basics

Beginner sample document | Learning and document Q&A; practice

1. What is machine learning?

Machine learning (ML) is a branch of artificial intelligence in which systems learn patterns from data to make predictions or decisions. A trained model applies learned patterns to new examples.

2. Features and targets

Features are the input variables supplied to a model. A target is the value a supervised model learns to predict. For house-price prediction, area and number of rooms may be features, while price is the target.

3. Main learning types

Supervised learning: learns from examples with known targets. Classification predicts categories, such as spam or not spam. Regression predicts numerical values, such as a house price.

Unsupervised learning: discovers structure without target labels. Clustering can group customers with similar behavior.

Reinforcement learning: an agent learns a policy through interaction with an environment and reward signals.

4. A typical ML workflow

Define the problem, collect data, inspect data quality, split the data, fit preprocessing on the training portion, train a model, tune using validation data and evaluate on held-out test data. Deploy and monitor when the results meet the application requirements.

Training data fits the model. Validation data guides model choices. Test data estimates performance after those choices are complete. Keep test data out of training and tuning.

5. Common algorithms

Linear regression predicts numerical values. Logistic regression is commonly used for classification despite its name. Decision trees split data using feature-based rules. Random forests combine multiple trees. K-means groups data into a chosen number of clusters.

6. Evaluating a model

Classification: accuracy measures the fraction of correct predictions. Precision measures how many predicted positives are truly positive. Recall measures how many actual positives are found. F1 is the harmonic mean of precision and recall.

Regression: mean absolute error (MAE) measures average absolute prediction error. Root mean squared error (RMSE) gives larger errors more weight. Lower MAE and RMSE indicate smaller prediction errors on the same dataset and scale.

7. Overfitting and data leakage

Overfitting occurs when a model learns training-specific details that do not generalize well. Underfitting occurs when a model fails to capture useful patterns. Cross-validation, regularization and suitable model complexity can help.

Data leakage occurs when information unavailable at prediction time enters training or evaluation. For example, fitting a scaler on the entire dataset before splitting can leak information from the test set.

8. Practice questions

What is a feature? How do classification and regression differ? Why separate training and test data? What is overfitting? Which metrics can evaluate a regression model?